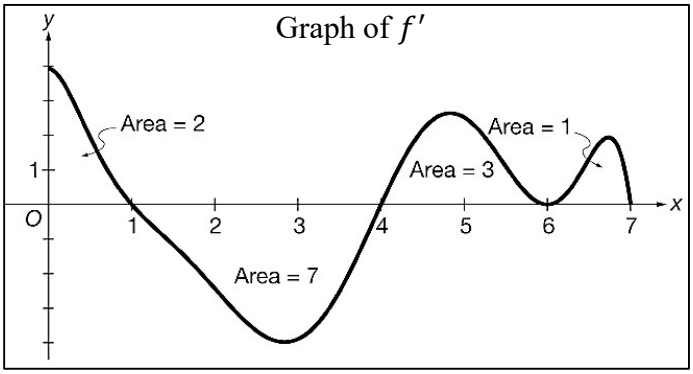
Unit 6 FRQ Practice Problems
AP Calculus AB Name_____

CALCULATORS ARE NOT ALLOWED ON THESE PROBLEMS! SHOW YOUR WORK.

1.) The figure at the right shows the graph of f', the derivative of a differentiable function f, on the closed interval $0 \le x \le 7$. The areas of the regions between the graph of f' and the x −axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(4) = 10$.



Let g be the function defined by $g(x) = 5 - x^2$.

(a) Find the value of $\int_0^7 f'(x)\,dx$.

(b) Given that $f(4) = 10$, write an expression for $f(x)$ that involves an integral. Use this expression to find the absolute minimum value of f and the absolute maximum value of f on the closed interval $0 \le x \le 7$. Justify your answers.

(c) Find $\int g(x)\,dx$.

(d) Find the value of $\int_1^2 x\,f'\big(g(x)\big)\,dx$.

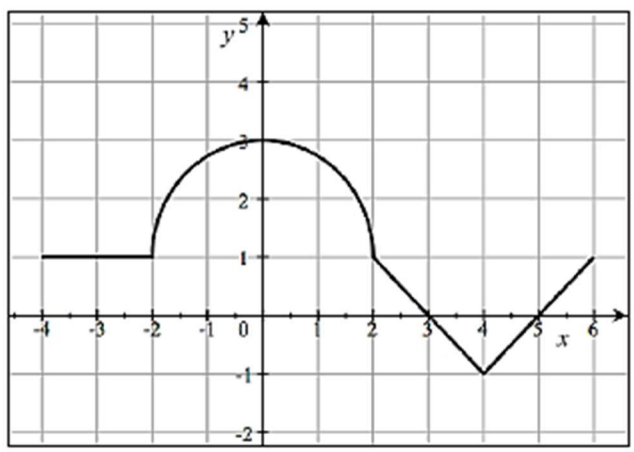
2.) Let f be a continuous function that is defined for all real numbers x and has the following properties:

(i) $\int_2^5 f(x)\,dx = \frac{-3}{4}$ (ii) $\int_2^8 f(x)\,dx = 3$

(a) Find $\int_8^5 f(x)\,dx$

(b) Find $\int_0^3 \left[2f(x+2) - x^2\right] dx$

3.) The graph of a function f consists of a semicircle and three line segments as shown below. Let g be the function given by $g(x) = \int_0^x f(t)\,dt$.



(a) Find $g(3)$.

(b) Find the equation of the tangent line to g at $x = 2$